

Data Visualization with Seaborn (Detailed Guide)

What is Seaborn?

Seaborn is a Python data visualization library built on top of **Matplotlib**. It helps create **beautiful, attractive, and statistical graphics** with less code.

Why Use Seaborn?

- ✓ Easy to create professional-looking charts
- ✓ Better default styles than Matplotlib
- ✓ Works directly with Pandas DataFrames
- ✓ Built-in statistical plots
- ✓ Less code compared to Matplotlib

Installation

```
pip install seaborn
```

Import Libraries

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

Load Sample Dataset

Seaborn provides built-in datasets.

```
tips = sns.load_dataset("tips")
print(tips.head())
```

Output

total_bill	tip	sex	smoker	day	time	size
16.99	1.01	Female	No	Sun	Dinner	2

1. Scatter Plot

Shows relationship between two numerical variables.

```
sns.scatterplot(x="total_bill", y="tip", data=tips)
plt.show()
```

Parameters

Parameter	Meaning
x	X-axis column
y	Y-axis column
data	Dataset

2. Line Plot

Shows trends over data.

```
sns.lineplot(x="size", y="tip", data=tips)
plt.show()
```

3. Bar Plot

Shows average values of categories.

```
sns.barplot(x="day", y="total_bill", data=tips)
plt.show()
```

Example Chart

4. Histogram

Shows frequency distribution.

```
sns.histplot(tips["total_bill"])
plt.show()
```

With Bins

```
sns.histplot(tips["total_bill"], bins=20)
plt.show()
```

5. KDE Plot

Shows smooth distribution curve.

```
sns.kdeplot(tips["total_bill"])  
plt.show()
```

6. Dist Plot (Old)

Deprecated but important for interviews.

```
sns.histplot(tips["total_bill"], kde=True)  
plt.show()
```

7. Box Plot

Used to find outliers.

```
sns.boxplot(x="day", y="total_bill", data=tips)  
plt.show()
```

Understanding Box Plot

- Middle line = Median
 - Box = Q1 to Q3
 - Whiskers = Data Range
 - Dots outside = Outliers
-

8. Violin Plot

Combination of Box Plot + Distribution.

```
sns.violinplot(x="day", y="total_bill", data=tips)  
plt.show()
```

9. Count Plot

Counts category frequency.

```
sns.countplot(x="day", data=tips)  
plt.show()
```

10. Pair Plot

Shows relationship between all numerical columns.

```
sns.pairplot(tips)
plt.show()
```

11. Heatmap

Shows data using colors.

```
corr = tips.corr(numeric_only=True)

sns.heatmap(corr, annot=True)
plt.show()
```

Parameters

Parameter	Meaning
-----------	---------

annot=True	Show values
------------	-------------

cmap	Color theme
------	-------------

Color Maps

```
sns.heatmap(corr, annot=True, cmap="coolwarm")
```

Popular colormaps:

- coolwarm
 - viridis
 - magma
 - YlGnBu
 - Blues
 - Reds
-

12. Correlation Matrix

```
tips.corr(numeric_only=True)
```

Heatmap Visualization:

```
sns.heatmap(
    tips.corr(numeric_only=True),
```

```
        annot=True,  
        cmap="viridis"  
    )  
plt.show()
```

13. Joint Plot

Scatter + Histogram

```
sns.jointplot(  
    x="total_bill",  
    y="tip",  
    data=tips  
)  
plt.show()
```

14. Regression Plot

Shows best-fit line.

```
sns.regplot(  
    x="total_bill",  
    y="tip",  
    data=tips  
)  
plt.show()
```

15. Strip Plot

Displays all data points.

```
sns.stripplot(  
    x="day",  
    y="total_bill",  
    data=tips  
)  
plt.show()
```

16. Swarm Plot

Better version of strip plot.

```
sns.swarmplot(  
    x="day",  
    y="total_bill",  
    data=tips  
)
```

```
plt.show()
```

17. Cat Plot

General categorical plot.

```
sns.catplot(  
    x="day",  
    y="total_bill",  
    kind="bar",  
    data=tips  
)  
plt.show()
```

Seaborn Themes

Dark Grid

```
sns.set_style("darkgrid")
```

White Grid

```
sns.set_style("whitegrid")
```

White

```
sns.set_style("white")
```

Dark

```
sns.set_style("dark")
```

Ticks

```
sns.set_style("ticks")
```

Change Figure Size

```
plt.figure(figsize=(10,5))  
sns.barplot(  
    x="day",  
    y="total_bill",  
    data=tips  
)  
plt.show()
```

Meaning

```
figsize=(10,5)
```

- 10 = Width
 - 5 = Height
 - Unit = Inches
-

Change Chart Color

```
sns.barplot(  
    x="day",  
    y="total_bill",  
    data=tips,  
    color="green"  
)
```

Add Title and Labels

```
sns.barplot(  
    x="day",  
    y="total_bill",  
    data=tips  
)  
  
plt.title("Average Bill by Day")  
plt.xlabel("Day")  
plt.ylabel("Total Bill")  
  
plt.show()
```

Difference Between Matplotlib and Seaborn

Feature	Matplotlib	Seaborn
Ease of Use	Medium	Easy
Styling	Basic	Beautiful
Statistical Plots	Limited	Excellent
Pandas Integration	Good	Excellent
Code Length	More	Less

Most Important Seaborn Functions

1. scatterplot()

2. `lineplot()`
3. `barplot()`
4. `histplot()`
5. `kdeplot()`
6. `boxplot()`
7. `violinplot()`
8. `countplot()`
9. `pairplot()`
10. `heatmap()`
11. `jointplot()`
12. `regplot()`
13. `stripplot()`
14. `swarmplot()`
15. `catplot()`

Interview Question

Q: What is the difference between Matplotlib and Seaborn?

Answer:

Seaborn is built on top of Matplotlib. It provides attractive statistical visualizations with less code and better default styles, while Matplotlib offers more customization and control.

Link : [CHATGPT](#)

Link : [ALL](#)